REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Conservation of Momentum Practice Questions

Grade 11 Advanced | Module 9 Lesson 2 | Conservation of Momentum

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: For negligible net external impulse: Σpi = Σpf. For zero net external torque: Li = Lf. Internal impulses cancel in system totals.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 11 book, printed pages 38–47.

P01. Total linear momentum is conserved when the system has:

A. No internal forces B. Zero kinetic energy
C. Negligible net external impulse D. Equal masses only

P02. In a two-object collision, the forces the objects exert on each other are:

A. Equal only if masses match B. Always in the same direction
C. Zero during contact D. Equal in magnitude and opposite in direction

---

P03. Recoil follows mainly from conservation of:

A. Momentum B. Temperature
C. Volume D. Density

P04. When two carts stick together, their final velocity is:

A. Always zero B. The same for both carts
C. Always the original faster speed D. Independent of their masses

---

P05. In the absence of net external torque, changing an object's shape can change:

A. Its total angular momentum necessarily B. The value of Planck's constant
C. Its angular velocity while angular momentum stays constant D. Its mass necessarily

---

P06. A 2 kg cart moving at +6.0 m/s collides with a stationary 4 kg cart. They stick together. Find their common velocity if external impulse is negligible.

A. 2 m/s B. 4 m/s
C. 1 m/s D. -2 m/s

P07. A stationary launcher of mass 4 kg fires a 0.2 kg projectile at +20 m/s. Find launcher recoil velocity if horizontal external impulse is negligible.

A. -2 m/s B. -1 m/s
C. -0.5 m/s D. 1 m/s

P08. Two fragments initially at rest separate. Fragment 1 has mass 4 kg and velocity +6.0 m/s; fragment 2 has mass 6 kg. Find fragment 2 velocity if external impulse is negligible.

A. -8 m/s B. -2 m/s
C. -4 m/s D. 4 m/s

P09. A skater changes moment of inertia from 4 to 2 kg m². Initial angular velocity is 3.0 rad/s. Find final angular velocity if net external torque is zero.

A. 12 rad/s B. 3 rad/s
C. -6 rad/s D. 6 rad/s

P10. A 4 kg cart moving +5.0 m/s collides with a 2 kg cart moving −2.0 m/s. Afterward the first moves +1.0 m/s. Find the second cart velocity; external impulse is negligible.

A. 6 m/s B. 12 m/s
C. 3 m/s D. -6 m/s

P11. Define an isolated system for a collision calculation.

P12. Why do internal collision forces not change the total momentum of the chosen two-object system?

P13. Can momentum be conserved while kinetic energy decreases?

P14. State conservation of angular momentum.

P15. Why must the system boundary be stated before applying momentum conservation?

P16. A 3 kg cart moving at +6.0 m/s collides with a stationary 6 kg cart. They stick together. Find their common velocity if external impulse is negligible.

P17. A 4 kg cart moving at +6.0 m/s collides with a stationary 8 kg cart. They stick together. Find their common velocity if external impulse is negligible.

P18. A stationary launcher of mass 6 kg fires a 0.3 kg projectile at +20 m/s. Find launcher recoil velocity if horizontal external impulse is negligible.

P19. A stationary launcher of mass 8 kg fires a 0.4 kg projectile at +20 m/s. Find launcher recoil velocity if horizontal external impulse is negligible.

P20. Two fragments initially at rest separate. Fragment 1 has mass 6 kg and velocity +6.0 m/s; fragment 2 has mass 9 kg. Find fragment 2 velocity if external impulse is negligible.

P21. Two fragments initially at rest separate. Fragment 1 has mass 8 kg and velocity +6.0 m/s; fragment 2 has mass 12 kg. Find fragment 2 velocity if external impulse is negligible.

P22. A skater changes moment of inertia from 6 to 3 kg m². Initial angular velocity is 3.0 rad/s. Find final angular velocity if net external torque is zero.

P23. A skater changes moment of inertia from 8 to 4 kg m². Initial angular velocity is 3.0 rad/s. Find final angular velocity if net external torque is zero.

P24. A 6 kg cart moving +5.0 m/s collides with a 3 kg cart moving −2.0 m/s. Afterward the first moves +1.0 m/s. Find the second cart velocity; external impulse is negligible.

P25. A 8 kg cart moving +5.0 m/s collides with a 4 kg cart moving −2.0 m/s. Afterward the first moves +1.0 m/s. Find the second cart velocity; external impulse is negligible.

P26. A 2.0 kg cart at +3.0 m/s sticks to a stationary 1.0 kg cart. Find final velocity and compare initial and final kinetic energy.

P27. Why can a rocket accelerate in space without pushing against air?

P28. Two skaters push apart from rest. One has twice the mass of the other. Compare their speed magnitudes.

P29. A rotating platform's moment of inertia halves without external torque. Predict angular speed and discuss kinetic energy.

P30. A collision on a rough surface is analysed by assuming exact momentum conservation for the carts alone. What condition must be checked?